SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Digital Signal Processing: DFT, FFT Algorithms, IIR & FIR Filter Design

Department: ECE	**Course / Subject:** Digital Signal Processing (EC501PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Discrete Fourier Transform (DFT), Radix-2 FFT algorithms (DIT and DIF), IIR filter design (Butterworth, Chebyshev), FIR filter design using Windows, and DSP processors.

Unit 1: Discrete Fourier Transform (DFT)

Properties of DFT, Circular convolution, Linear convolution using DFT, Circular correlation, and Fast convolution methods (Overlap-Add, Overlap-Save).

Unit 2: Fast Fourier Transform (FFT) Algorithms

Radix-2 Decimation-In-Time (DIT) FFT, Decimation-In-Frequency (DIF) FFT, Butterfly signal flow graphs, and Computational complexity reduction from O(N^2) to O(N log N).

Unit 3: IIR Digital Filter Design

Analog filter approximations (Butterworth, Chebyshev Type I), Impulse Invariance Method, Bilinear Transformation Method (BTM), Warping effect, and Pre-warping.

Unit 4: FIR Digital Filter Design

Linear phase conditions of FIR filters, Symmetric and anti-symmetric impulse response, Windowing design techniques (Rectangular, Hamming, Hanning, Blackman).

Unit 5: Digital Filter Realization & Quantization

Direct Form I and II, Cascade, and Parallel realizations of IIR systems, Transversal and Linear phase FIR realization, Finite word-length quantization effects.